INCH-POUND

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DETAIL SPECIFICATION

HOSE ASSEMBLIES, POLYTETRAFLUOROETHYLENE, ARAMID FIBER REINFORCED, 5000 AND 8000 PSI, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements for polytetrafluoroethylene (PTFE) lined, aramid fiber reinforced, hose assemblies with dynamic beam seal end fittings for use in aerospace fluid systems.

1.2 Classification. Hose assemblies are of the following type and service.

1.2.1 Type. Type II, -65 ºF to +275 ºF temperature range.

1.2.2 Service. For service to 5000 psi, the hose assembly is suitable for use at the small bend radius. For service above 5000 psi, the hose assembly must be used with the larger bend radius listed in table II.

1.2.3 Part or Identifying Number (PIN). The PIN is as specified in the applicable specification sheet.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has

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been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specification, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680	-	Degreasing Solvent
MIL-PRF-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-87257	-	Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

(See ASSIST database for a list of specification sheets.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-31000	-	Technical Data Packages

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1	-	Surface Texture, (Surface Roughness, Waviness and Lay)

(Copies of these documents are available at: https://www.asme.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) INTERNATIONAL

ASTM A262	-	Standard Practices for Detecting Susceptibility to

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		Intergranular Attack in Austenitic Stainless Steels
ASTM D380	-	Standard Test Methods For Rubber Hose
ASTM D792	-	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement

(Copies of these documents are available at www.astm.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AMS3380	-	Hose, Polytetrafluoroethylene, TFE Fluorocarbon Resin, Wire Braid Reinforced
SAE AMS4944	-	Titanium Alloy Tubing, Seamless, Hydraulic 3.0A1-2.5V Cold Worked, Stress Relieved
SAE AMS5561	-	Steel, Corrosion and Heat Resistant, Welded and Drawn or Seamless and Drawn Tubing, 9.0Mn-20Cr-6.5Ni-0.28N High-Pressure Hydraulic
SAE AS603	-	Impulse Testing of Hydraulic Hose, Tubing and Fitting Assemblies
SAE AS611	-	Hose Assembly and Tubing, Polytetrafluoroethylene, Cleaning Methods for
SAE AS2078	-	Test Methods, Hose Assemblies, Polytetrafluoroethylene (PTFE)
SAE AS5272	-	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting, Procurement Specification
SAE AS8879	-	Screw Threads – UNJ Profile, Inch, Controlled Radius Root With Increased Minor Diameter
SAE AS85720	-	Fittings, Tube, Fluid Systems, Separable, High Pressure Beam Seal, 5000/8000 psi, General Specification for
SAE AS85720/1	-	Fittings, Tube, Fluid Systems, Separable, High Pressure, Dynamic Beam Seal, Design Standard for Male End

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of document and the references cited herein, the text of this document

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takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.3.

3.3 Material. The hose assembly materials shall be uniform in quality, free from defects, consistent with good manufacturing practice and shall conform to applicable specifications and requirements specified herein. All materials not specifically described herein shall be of the highest quality and suitable for the purpose intended.

3.3.1 Fittings. The material used shall be titanium alloys or corrosion resistant steels as specified in SAE AS85720. The fitting sockets shall be corrosion resistant steel. The material shall be heat treated, as required, and have suitable surface protection to meet requirements of this specification.

3.3.2 Sockets. Crimped or swaged sockets of 304 steel shall pass testing in accordance with ASTM A262 practice E prior to crimping or swaging. This requirement does not apply to sockets made from 321, 347 or 304L corrosion resistant steel.

3.3.3 Tubing. The tubing used as an integral part of the hose end fittings shall be in accordance with AMS 4944 or AMS 5561.

3.4 Design and construction. The hose assembly shall consist of a constructed seamless inner tube of virgin polytetrafluoroethylene (PTFE), reinforced with aramid fiber, suitably treated, a black cover and end fittings. The cover shall be colored black to protect the aramid fiber from ultraviolet light and chafing. The aramid fiber shall be a contrasting color to the black cover to allow for the inspection of 100 percent coverage.

3.4.1 Styles and dimensions. The styles and dimensions shall conform to the design requirements specified herein and applicable drawings as approved by the procuring activity.

3.4.2 Inner tube. The inner tube shall be of seamless construction of uniform gage. It shall have a smooth bore and shall be free from pitting or projections on the inner surface. Additives may be included in the compound from which the tube is extruded.

3.4.3 Reinforcement. The reinforcement shall consist of an aramid braid. The braid shall be so arranged over the inner tube as to provide sufficient strength to insure conformance with the requirements specified herein. Broken reinforcing cords are cause for rejection.

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3.4.4 Outer protective cover. The aramid fiber reinforcement shall be adequately protected from chafing and long term exposure from ultraviolet rays with the addition of a braided outer protective cover. The outer cover shall be completely entrapped within the fitting socket.

3.4.5 Fittings. The end fittings for the hose assembly shall meet the requirements of SAE AS85720 and SAE AS85720/1. The materials shall conform to the applicable specifications listed herein. The outlet design shall be dynamic beam seal type and meet the requirements of SAE AS85720.

3.4.5.1 Screw threads. All threads shall be in accordance with SAE AS8879, with the surface finish of the thread flanks being 62 Ra maximum. All threads shall be coated with solid film lubricant in accordance with SAE AS5272, type I or type II.

3.4.6 Fluid media. The hose assembly shall be compatible both internally and externally with MIL-PRF-83282, MIL-PRF-87257 and MIL-PRF-5606.

3.4.7 Weight. The hose assembly shall be the lightest weight practical consistent with good design practice that is capable of meeting the requirements of this specification.

3.4.8 Length. Hose assembly lengths and tolerances shall be as specified in the applicable specification sheet.

3.5 Performance.

3.5.1 Inner tube.

3.5.1.1 Tube roll. The tube, without reinforcement braid, shall not leak, split, burst, or show any other evidence of malfunction, when rolled through the sequence of rollers as specified in 4.5.2.1.

3.5.1.2 Tube proof pressure. The tube, without reinforcement braid, shall not leak, burst, or show any evidence of malfunction when held for a minimum of one minute at the proof pressure specified in table I.

3.5.1.3 Tube tensile strength. The longitudinal tensile strength for all sizes of tubes shall be 2200 psi minimum. The transverse tensile strength for sizes -10 and larger shall be 1800 psi (12500 kPa) minimum. For sizes -08 and smaller, the transverse tensile strength need not be tested. The test method is specified in 4.5.2.2.

3.5.1.4 Specific gravity. The specific gravity values of the hose inner tube shall conform to the specific gravity values as required herein, when tested as specified in 4.5.2.3.

3.5.1.5 Elongation. The elongation of PTFE tube at 77 ±2 ºF shall be in accordance with the values specified in 4.5.3.2.

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TABLE I. Tube proof pressure.

Size dash no.	Flattening gap max. (inches)	Rounding gap min. (inches)	Proof pressure (psi)
-03	.281	.250	380
-04	.281	.250	380
-06	.281	.328	280
-08	.328	.469	220
-10	.328	.578	170
-12	.328	.688	130
-14	.328	.765	110
-16	.328	.828	95

3.5.2 Hose assembly. The hose assembly (complete with reinforcing braid, cover and assembled with fittings) shall meet the following performance requirements.

3.5.2.1 Proof pressure. When tested as specified in 4.5.3.1, the hose assembly shall withstand the proof pressure listed in table II without malfunction or leakage.

3.5.2.2 Elongation and contraction. The hose assembly shall not change in length by more than ±0.30 inch in 10 inches of length, when subjected to the operating pressure shown in table II for a minimum of five minutes. The hose assembly shall be tested in accordance with 4.5.3.2.

3.5.2.3 Volumetric expansion. The volumetric expansion of the hose assembly shall not exceed the limits specified in table II, when tested in accordance with 4.5.3.3.

3.5.2.4 Leakage. The hose assembly shall not leak when subjected to two pressure cycles of 67 percent of minimum burst pressure in table II, when tested in accordance with 4.5.3.4.

3.5.2.5 Room temperature burst pressure. The hose assembly shall not leak nor burst at any pressure below the room temperature burst values specified in table II, when tested in accordance with 4.5.3.5.

3.5.2.6 High temperature burst pressure. The hose assembly shall not leak nor burst at any pressure below the high temperature burst value specified in table II, when tested in accordance with 4.5.3.6.

3.5.2.7 Thermal shock. The hose assembly shall not leak nor show any evidence of malfunction when exposed to thermal shock of -65 ºF to +275 ºF, when tested in accordance with 4.5.3.7.

3.5.2.8 Impulse. The hose assembly shall be capable of withstanding 200,000 impulse cycles at +275 ºF, when tested in accordance with 4.5.3.8. Testing at both 5000 psi and 8000 psi with the minimum bend radii as specified in table II is required.

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TABLE II. Physical requirements of high pressure hose assemblies.

Size dash no.	Operating pressure (max. psi)	Proof pressure (min. psi)	Burst pressure room temp (min. psi)	Burst pressure hi temp. (min. psi)	Min. bend radius (inside of bend - Inches)		Volumetric expansion (cc/in.)
					Operating pressure		
					5000 psi	8000 psi	
-03	8000	16,000	28,000	24,000	1.25	2.50	.09
-04	8000	16,000	28,000	24,000	1.50	3.00	.13
-06	8000	16,000	28,000	24,000	2.50	5.00	.21
-08	8000	16,000	28,000	24,000	2.88	5.75	.35
-10	8000	16,000	28,000	24,000	3.25	6.50	.55
-12	8000	16,000	26,000	22,000	4.00	7.75	.75
-14	8000	16,000	25,000	21,000	4.50	8.75	1.20
-16	8000	16,000	24,000	20,000	5.00	9.63	1.60

3.5.2.9 Assembly flexibility. The hose assembly shall not leak nor show any evidence of malfunction when flex-cycled from -65 to +275 ºF, when tested in accordance with 4.5.3.9. Testing at both 5000 psi and 8000 psi with the minimum bend radii as specified in table II is required.

3.5.2.10 Stress degradation. The hose assembly shall not exceed the air leakage as specified, when tested in accordance with 4.5.3.10.

3.5.2.11 Pneumatic surge. The inner tube of the hose assembly shall not collapse or show evidence of degradation, when tested in accordance with 4.5.3.11.

3.5.2.12 Pneumatic effusion. The hose assemblies, when tested in accordance with 4.5.3.12, shall not exceed the effusion rate specified herein.

3.5.2.13 Overtightening torque. The fitting shall withstand the overtightening torque values specified in table III, when tested in accordance with 4.5.3.13.

3.5.2.14 Conductivity. When tested as specified in 4.5.3.14, the hose assembly shall conduct a direct current equal to 900 microampere minimum to 10000 microampere maximum with a test potential of 1000 VDC.

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TABLE III. Overtightening torque values.

Fitting size	Inch-pounds
-03	130
-04	202
-06	360
-08	576
-10	792
-12	1008
-14	1224
-16	1656

3.6 Item marking.

3.6.1 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-31000 shall govern the manufacturer's part numbers and changes thereto.

3.6.2 Identification product. Hose assemblies shall be marked for identification in accordance with MIL-STD-130. The following special markings shall be added:

a. Fittings: The manufacturer's name or trademark shall be permanently marked on all end fittings.

b. Assembly: A permanent marking on the fitting or a permanent band on the assemblies shall be used. The band shall be retained onto the fitting socket and shall not contact the hose cover. The band shall be no wider than 1 inch and shall not impair the flexibility or the performance of the hose or damage the outer fibers. The marking on the fitting or band shall include the following information:

(1) Assembly manufacturer's name or trademark and assembly specification "MIL-DTL-85800."

(2) Hose manufacturer's Federal Code number.

(3) Operating pressure "8000 psi"

(4) Operating temperature " 275 ºF"

(5) Pressure test symbol "PT."

(6) Date of hose assembly manufacture expressed in terms of month and year.

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3.7 Workmanship. Remove all burrs and break all sharp edges. Sealing surfaces shall be free of detrimental longitudinal and spiral tool marks. Unless a finer finish is specified in applicable drawings, sealing surfaces shall be smooth to a finish of 45 Ra, except the annular tool marks up to 45 Ra, as defined in ASME B46.1 will be acceptable. All other machined surfaces shall be smooth to 125 Ra maximum and free of burrs and sharp edges. Unmachined surfaces, such as forging surfaces and bar stock flats, shall be free of cracks, laps and seams.

3.8 Internal cleanliness. Prior to shipping, the end fittings of the hose assembly shall be capped or plugged to prevent entrance of moisture and foreign matter. The caps or plugs shall be securely attached and shall withstand normal strains, jarring, vibrations encountered during shipping, storage and handling. The interior surface of the hose assembly shall be free from oil, grease, dirt, moisture, cleaning solvents and foreign materials. All hose assemblies shall be cleaned in accordance with SAE AS611, class 0. During conformance inspection, hose assemblies with uncovered ends shall be rejected and considered as a failure. The interior of the assembly shall not contain any debris or foreign materials when examined in accordance with 4.5.3.15.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions.

4.2.1 Fitting ends. First article and quality conformance inspection shall be conducted on assemblies using straight type swivel ends, except for assemblies for impulse test shall have a 90-degree swivel on one end.

4.2.2 Preparation of specimen.

4.2.2.1 Sample lengths. Unless otherwise specified, length of sample hose assemblies shall be in accordance with table IV.

4.2.2.2 Oil aging. In all the tests using oil aged samples, the hose assemblies shall be fully immersed in MIL-PRF-83282 hydraulic fluid and soaked in an air oven at a temperature of 275 ºF for seven days. All air shall be excluded from the bore of the assembly during the test. No pressure shall be applied to the assembly during the aging period.

4.2.2.3 Air aging. Air-aged samples shall be kept in air at a temperature of +275 ºF for seven days.

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4.2.2.4 Unaged assemblies. Unaged assemblies shall be as shipped from the hose assembly manufacturer.

TABLE IV. Length of hose assemblies for test (in inches).

Hose assembly	Twelve assemblies for impulse test, six for each pressure		Four assemblies for flex test	Six assemblies for other tests 1/
	Operating pressure			
	5000 psi	8000 psi		
-03	11	15	18	18
-04	12	16	20	18
-06	15	21	27	18
-08	18	24	30	18
-10	21	30	33	18
-12	25	33	37	18
-14	28	37	41	18
-16	31	41	45	18

1/ One additional sample of each size in lengths as shown on figure 2 shall be used for examination and conductivity tests.

4.2.3 Test conditions.

4.2.3.1 Test fluids. Unless otherwise specified, the pressure test fluid shall be hydraulic oil conforming to MIL-PRF-83282, MIL-PRF-87257 or other suitable fluid.

4.2.3.2 Test pressures. Inner tube proof pressures are specified in table I. Hose assembly test pressures are specified in table II. Unless otherwise specified, all hose assembly test pressures shall have a tolerance of ±100 psi. Unless otherwise specified, all performance tests in 4.5 shall be conducted with an operating pressure of 8000 psi.

4.2.3.3 Temperature. Except where otherwise specified, the tests shall be conducted at a room temperature of approximately +60 ºF to +100 ºF and a fluid temperature of +60 ºF to +170 ºF. Unless otherwise specified, temperature measurements shall be taken within six inches of the hose assemblies under test. Except as otherwise noted, all temperatures shall have a tolerance of ±15 ºF.

4.2.3.4 Torque. One end of each assembly shall be torqued to the minimum torque value and the other end torqued to the maximum torque value without application of any additional lubricant beyond that supplied by the manufacturer. Torque values for end fittings are specified in SAE AS85720.

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4.3 First article inspection. First article inspection shall consist of all tests listed in table V. Test sequence shall be as shown in table VI.

4.3.1 First article test samples. Test samples shall consist of the quantity and length specified in table IV. The samples shall be capable of mating with the dynamic beam seal type design, in accordance with SAE AS85720. All hose assembly samples shall be representative of the construction, workmanship, components and materials used in production. When a manufacturer is in continuous production of hose or hose assemblies from one contract to another, submission of additional first article samples for a new contract may be waived at the discretion of the acquiring activity (see 6.2).

4.3.2 Failures. One or more failures shall be cause for refusal to grant first article approval.

TABLE V. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product	3.3, 3.4, 3.6 and 3.7	4.5.1
Tube roll and proof test	3.5.1.1, 3.5.1.2	4.5.2.1
Tube tensile strength	3.5.1.3	4.5.2.2
Proof pressure test	3.5.2.1	4.5.3.1
Elongation and contraction test	3.5.2.2	4.5.3.2
Volumetric expansion test	3.5.2.3	4.5.3.3
Leakage test	3.5.2.4	4.5.3.4
Room temperature burst pressure test	3.5.2.5	4.5.3.5
High temperature burst pressure test	3.5.2.6	4.5.3.6
Thermal shock test	3.5.2.7	4.5.3.7
Impulse test	3.5.2.8	4.5.3.8
Assembly flex test	3.5.2.9	4.5.3.9
Stress degradation test	3.5.2.10	4.5.3.10
Pneumatic surge test	3.5.2.11	4.5.3.11
Pneumatic effusion test	3.5.2.12	4.5.3.12
Overtightening torque test	3.5.2.13	4.5.3.13
Conductivity test	3.5.2.14	4.5.3.14

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TABLE VI. First article inspection test sequence.

Assembly numbers	1	2 through 5	6 and 7	8 and 9	10 and 11	12 through 24 1/	25
Inspection sequence	4.5.1 4.5.2.1 4.5.2.2	4.5.1 4.5.3.1 4.5.3.2 4.5.3.9 4.5.3.13 4.5.3.5	4.5.1 4.5.3.12 4.5.3.1 4.5.3.3 4.5.3.6	4.5.1 4.5.3.1 4.5.3.4 4.5.3.7	4.5.1 4.5.3.1 4.5.3.10 4.5.3.11	4.5.1 4.5.3.1 4.5.3.8	4.5.1 4.5.3.1 4.5.3.14

1/ These samples shall have a 90º elbow fitting on one end of the hose, and a straight-type fitting on the other end of the hose. If approval is being sought for both the bent-tube and the forged elbow configuration then one-half of the samples (6) shall use one type of configured elbow while the other half of the samples use the other type.

4.4 Quality conformance inspections. Quality conformance inspections shall be accomplished on hose assemblies for delivery in accordance with 4.4a, 4.4b and 4.4c.

a. Individual tests (see 4.4.1) (100 percent inspection).

b. Sampling tests (see 4.4.2).

c. Periodic controls (see 4.4.3).

4.4.1 Individual tests. Each hose assembly shall be subjected to the following tests:

a. Examination of product of 4.5.1.

b. Proof pressure test of 4.5.3.1.

NOTE: Production samples that are proof pressure tested with water should be air dried prior to capping.

4.4.2 Sampling tests. The following inspections and tests shall be performed on eight hose assemblies with straight fittings on each end, selected at random from each inspection lot. Initially, an inspection lot shall be hose made from each of the first four reinforcement setups for the construction of each dash size. After successfully testing four consecutive reinforcement setups (lots), an inspection lot shall consist of no more than 3000 hose assemblies, all of one hose size, manufactured under essentially the same conditions, but not necessarily during one continuous run. One hose assembly tested for each lot of 375 hose assemblies is also permitted for protracted or small assembly runs.

a. Internal cleanliness test of 4.5.3.15.
b. Leakage test of 4.5.3.4.

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c. Room temperature burst test of 4.5.3.5.

4.4.3 Periodic control tests. The following tests shall be performed on eight hose assemblies (for each test) individually selected at random from each complete lot. A periodic control test lot shall consist of no more than 20,000 feet of hose, all of one dash size, manufactured under essentially the same conditions, but not necessarily during one continuous run. Two hose assemblies may be tested for each lot of 5000 feet is also permitted

a. Stress degradation test of 4.5.3.10 (samples 5 through 8).

b. Conductivity test of 4.5.3.14 (samples 5 through 8).

c. Impulse test of 4.5.3.8 (oil aged samples only, samples 1 through 4).

d. Elongation and contraction test of 4.5.3.2. (samples 1 through 4).

4.4.4 Rejection and retest. All inspection plans shall be single sample plans with an acceptance number of zero. Parts subjected to non-destructive tests and failing to conform to the requirements of these tests shall be rejected. Parts subjected to destructive tests shall be discarded. When a hose assembly fails to pass a sampling test, the entire lot represented by the sample shall be rejected.

4.4.7 Destructive test sample. Prior to testing, a letter (D) shall be impression stamped on each end fitting of hose assemblies used for destructive tests of 4.4.2 and 4.4.3.

4.5 Method of examination and test.

4.5.1 Examination of product. . Each hose assembly shall be visually and physically examined for conformance to the configuration specified in the applicable specification sheet and the following requirements:

a. Materials (see 3.3)
b. Design and construction (see 3.4)
c. Item marking (see 3.6)
d. Workmanship (see 3.7)

4.5.2 Inner tube tests.

4.5.2.1 Tube roll and proof test. Each length of tubing shall be subjected to a tube roll and proof test in accordance with AMS 3380, except that the flattening gap, rounding gap and proof pressure shall be as specified in table I. The test fluid shall be either air or water.

4.5.2.2 Tube tensile strength. The test samples from each lot of tube shall be subjected to tensile strength tests in accordance with AS2078.

4.5.2.3 Specific gravity of tube.

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4.5.2.3.1 Apparent specific gravity. Apparent specific gravity shall be determined in accordance with ASTM D792, Method A, and shall not exceed 2.155 at 77 ºF ±2 ºF. Two drops of wetting agent shall be added to the water. When test samples are prepared from braided hose, the braid impressions must be removed before testing.

4.5.2.3.2 Relative specific gravity. Relative specific gravity shall not exceed a value of 2.190 for all sizes and types of tubes.

4.5.3 Hose assembly tests.

4.5.3.1 Proof pressure test. All hose assemblies shall be pressure tested to the values specified in table II for not less than 30 seconds and not more than five minutes. Any evidence of leakage from hose or fittings or any other evidence of malfunction shall constitute failure.

4.5.3.2 Elongation and contraction test. Two hose assemblies of each size shall be subjected to the elongation and contraction test. The hose shall not change in length by more than ±0.30 inch in 10 inches of length when subjected to the operating pressure shown in table II for not less than five minutes. With the hose held in a straight, unpressurized condition, a 10-inch gage length shall be marked off on the hose and the hose then pressurized. After five minutes, while still pressurized, the gage length shall be measured and that change in length be calculated.

4.5.3.3 Volumetric expansion test. Two hose assemblies of each size shall be tested in accordance with ASTM D380. The volumetric expansion of the test assemblies shall be in accordance with the values shown in table II. This test shall be performed at the operating pressure shown in table II.

4.5.3.4 Leakage test. Two hose assemblies of each size shall be pressurized to 67 percent of the minimum room temperature burst pressure shown in table II, and held for five minutes minimum. The pressure shall then be reduced to zero psi, after which it shall be raised to 67 percent of the minimum room temperature burst pressure for a final 5-minute check. Any evidence of leakage from hose or fittings, hose burst, fitting blow-off or any other evidence of malfunction shall constitute a failure.

4.5.3.5 Room temperature burst pressure test. Two hose assemblies of each size shall be oil aged in accordance with 4.2.2.2, then they shall be subjected to a pressure sufficient to burst the assemblies with a rate of pressure rise equal to 20,000 psi ±5000 psi per minute. The hose assemblies shall be observed throughout the test and the type of failure and the pressure at which failure occurred shall be recorded. The assemblies shall not leak or show any evidence of malfunction at any pressure below the specified pressure listed in table II.

4.5.3.6 High temperature burst pressure test. Two hose assemblies of each size shall be oil aged in accordance with 4.2.2.2, then they shall be filled with MIL-PRF-83282 fluid, placed into a suitable container and then into an oven preheated to 275 ºF. There, the hose assemblies shall be soaked for one hour with ambient and fluid temperature at 275 ºF. At the end of that period, the assemblies shall be pressurized to proof pressure of table II for a minimum of five minutes.

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The pressure shall then be released and while maintaining at 275 ºF, the pressure shall be increased to failure as described in 4.5.3.5.

4.5.3.7 Thermal shock test. The thermal shock test shall be as follows:

a. Two hose assemblies of each size be subjected to this test. One assembly shall be air-aged and one assembly shall be unaged. The assemblies shall be subjected to the proof pressure specified in table II for a minimum of five minutes.

b. The test assemblies shall then be mounted, empty, in a high temperature test setup (typical setup shown on figure 1) and the ambient temperature reduced to -65 ±2 ºF for a minimum of two hours. At the end of this period, while still at this temperature, high temperature test fluid at a temperature of +275 ºF shall be suddenly introduced at a minimum pressure of 50 psi. Immediately after the hot oil has filled the assembly, the pressure shall be raised to the proof pressure specified in table II for a minimum of five minutes. Not more than 15 seconds shall elapse between the introduction of the high temperature oil at 50 psi and the raising of the pressure to proof pressure.

c. The test assemblies shall then be filled with one of the high temperature test fluids at a pressure of 75 ±25 psi and soaked with fluid and ambient temperature maintained at 275 ºF for one hour. At the end of this period, the assembly shall be pressurized to proof pressure specified in table II for a minimum of five minutes. The pressure shall be released and while still maintaining at 275 ºF, the pressure shall be increased at the same rate of rise as specified in 4.5.3.5 until failure is obtained. The hose assembly shall be under continuous observation during the preceding test and the pressure at which failure occurred and the type of failure shall be recorded.

d. During part b and the proof portion of part c of the test, any evidence of leakage from the hose or fittings, hose burst, fitting blow-off or any other evidence of malfunction shall constitute failure. During the burst portion of c, any of the above occurring below the minimum high temperature burst pressure shown in table II shall constitute a failure.

4.5.3.8 Impulse test. The impulse test shall be as follows:

a. Four test assemblies of each size shall be oil-aged, four shall be air-aged, and four shall be unaged. The assemblies shall then be subjected at room temperature to a proof pressure specified in table II for a minimum of five minutes.

b. Six samples, two each oil-aged, air-aged and unaged shall be tested with a 5000 psi operating pressure, 7500 psi peak pressure and 5000 psi bend radius as specified in table II. Six samples, two each oil-aged, air-aged and unaged shall be tested with an 8000 psi operating pressure, 10,000 psi peak pressure and 8000 psi bend radius as specified in table II. The impulse pattern shall be as specified in SAE AS603.

c. Electronic measuring devices shall be used to determine and control the impulse pressure in the inlet manifold. The impulse shall occur at 70 ± 10 cycles per minute (cpm). The

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test fluid and ambient air shall be maintained at +275 ºF and measured at the test manifold.

The test shall be run in such a manner that the assemblies shall be temperature cycled from room temperature to specified ambient and fluid temperatures for 200,000 impulse cycles with 80% of the cycles at +275 ºF and 20 percent at room temperature. Any evidence of leakage from the hose or fittings prior to the completion of 200,000 impulse cycles for all sizes shall constitute failure.

4.5.3.9 Assembly flexibility test. Four hose assemblies of each size shall be mounted in the assembly flexure test setup as illustrated on figure 2 and subjected to the following test sequence. Two hose assemblies of each size shall be tested to 5000 psi pressure requirements with the "A" dimension and 5000 psi minimum bend radii as specified herein. Two assemblies of each size shall be tested to 8000 psi pressure requirements with the "A" dimension and 8000 psi minimum bend radii as specified herein. The assemblies shall be filled with one of the high temperature test fluids as specified in 4.2.3.1. Temperature indicated is both fluid and ambient. Flexing shall occur at a rate of 70 ±10 cpm during portions c, d, and e.

a. The test assemblies shall be soaked with no pressure or flexing at a temperature of -65 ± 2 ºF for a minimum of one hour.

b. With no flexing, the test assemblies shall be pressurized to the proof pressure as specified in table II with the temperature still at -65 ºF for a minimum of five minutes (first cycle only).

c. Flexing shall begin while the test assemblies are pressurized to the operating pressure as specified in table II with the temperature still at -65 ºF for a minimum of 4000 cycles.

d. With the pressure reduced to zero psi, flexing shall continue for 1000 cycles at -65 ºF.

e. Increase the temperature to +275°F and flex 1000 cycles with the pressure at zero psi. The pressure shall then be increased to the operating pressure as specified in Table II with the temperature held at +275°F. Flexing shall continue until an accumulated total of 80,000 cycles is reached.

f. Steps (a), (c), (d) and (e) shall be repeated for a total of five test sequences, i.e., 400,000 flexing cycles.

g. After completion of step (f) and with no flexing, the test assemblies shall be pressurized to the proof pressure specified in Table II with the temperature still at +275°F for a minimum of five minutes (last cycle only).

Any leakage from the hose or fittings, hose burst, fitting blow-off or any other evidence of malfunction during the test shall constitute failure.

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4.5.3.10 Stress degradation test. Two hose assemblies of each size shall be subjected to this test.

a. The assemblies shall be filled with test fluid and then placed in an oven which shall be maintained at a temperature of +275 ºF. Precautions shall be taken to assure that the hose assemblies do not come in contact with parts of the oven that are at a higher temperature. A pressure equal to the rated operating pressure specified in table II shall be applied to the hose assemblies.

b. After a minimum of 20 hours at +275 ºF, the pressure shall be gradually released and the assemblies shall be removed from the oven, drained and cooled to room temperature.

c. The hose assemblies shall then be filled with test fluid and a pressure equal to the rated operating pressure specified in table II shall be applied and held for a minimum of two hours at room temperature.

d. The procedure specified in (a), (b) and (c) shall be repeated for a total of three times.

e. After the final two-hour pressurization period, the hose assemblies shall be drained and flushed with solvent conforming to type 1 of MIL-PRF-680.

f. The hose assemblies shall be air dried and then subjected to a pneumatic effusion test in accordance with 4.5.3.12. To conduct this test, the hose shall be installed in an apparatus similar to that shown on figure 4.

g. The apparatus with the hose assembly installed shall be immersed in water containing no wetting agent. A pressure equivalent to the rated operating pressure specified in table II shall be applied for 15 minutes to allow any entrapped air in the hose to escape.

h. The pressure shall be held an additional five-minute period during which time the effused gas shall be collected from the test sample, including the juncture of the hose to the fitting, but not including the "B" nut. After the five-minute period of pressurization, the average rate of effusion through the hose and two fittings shall be computed to cc/in./min. If the average rate of effusion exceeds 2.0 cc/in./min for any size, it shall be cause for rejection.

4.5.3.11 Pneumatic surge test. Two hose assemblies that were subjected to the stress degradation test (see 4.5.3.10) shall be used for this test. The hose assemblies shall be installed in the test apparatus in accordance with figure 3. The assemblies shall be subjected to the rated operating pressure as specified in table II for 25 minutes at room temperature. After this period of pressurization, the exhaust valve shall be opened within 50 milliseconds to permit rapid discharge of the compressed air or nitrogen gas. After five minutes, the valve shall be closed and the pressure recycled. This sequence of 25 minutes at operating pressure and five minutes at zero pressure shall be repeated at total of 16 times. At the end of this period, the hose shall be sectioned and examined for evidence of tube collapse, sponging of inner tube, etc., and the filter

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downstream of the hose examined for evidence of inner tube degradation. Any evidence of degradation shall constitute a failure.

4.5.3.12 Pneumatic effusion test. Two hose assemblies of each size shall be used for this test. The assemblies shall be subjected to the operating pressure specified in table II for one hour. Air effusion shall be collected using the water displacement method and air collecting device similar to that depicted on figure 4. The total amount of effusion through the hose and two fittings shall be collected over the last ½ hour of testing. Total effusion shall not exceed 8.0 cc/ft of hose assembly for any size hose.

4.5.3.13 Overtightening torque test. Two hose assemblies with straight swivel ends shall be subjected to this test. Prior to this test, the fittings shall be lubricated with oil (see 4.2.3.1). The fittings shall be tightened to the applicable torque specified in table III and then loosened. This sequence shall be repeated 15 times. After this sequence, there shall be no evidence of failure or deformation of the fitting assemblies, and the swivel nuts shall be free enough to permit turning on the nipple by hand. After the above tightening torque test sequence, the tested end fittings shall be coupled to a length of hose and subjected to the proof pressure specified in table II for five minutes without leakage.

4.5.3.14 Conductivity test. The test shall be performed in accordance with the conductivity test for non-metallic braided hose and hose assembly specified in SAE AS2078.

4.5.3.15 Internal cleanliness. Both ends of the hose assembly shall be visually inspected to determine if caps or plugs are installed at the fittings. For hose assembly with straight fittings at both ends, remove the caps or plugs and place a light source at one end of the hose assembly. The interior of the hose shall be visually examined, without magnification, from the opposite end of the light source. Conformance shall be as specified in 3.8.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

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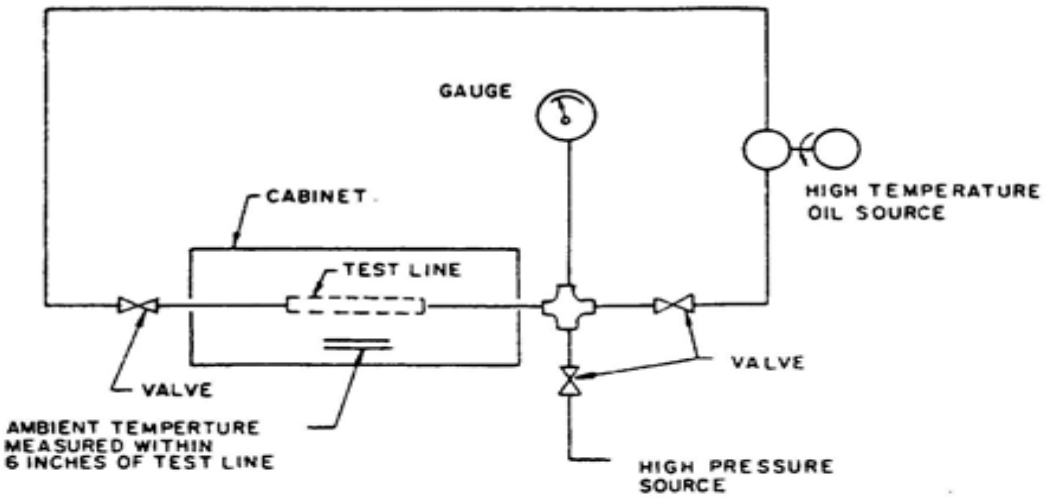
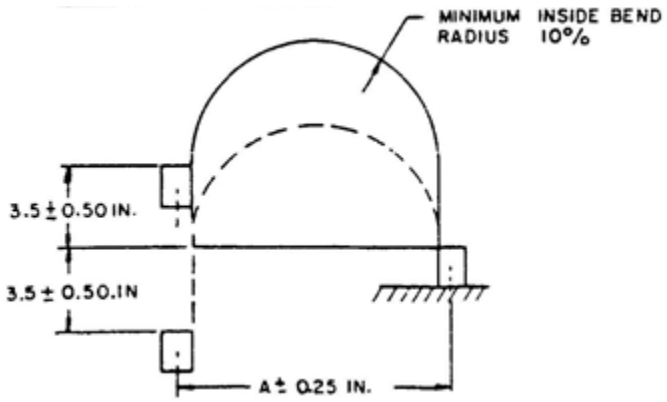


FIGURE 1. Typical setup for thermal shock testing.



Hose size Dash number	A (inches)	
	Operating pressure	
	5000 psi	8000 psi
-03	3.00	6.00
-04	3.50	6.50
-06	5.62	10.62
-08	6.50	12.25
-10	7.25	14.00
-12	9.00	16.62
-14	10.25	18.50
-16	11.12	20.75

FIGURE 2. Assembly flexure test setup.

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6. NOTES

(This section contains information of a general or explanatory nature, that may be helpful, but is not mandatory.)

6.1 Intended use. The hose assemblies covered by this specification are intended for use in aircraft high pressure 5000 and 8000 psi hydraulic systems operating in a temperature range of -65º to +275 ºF. Do not use in fire zones or a pump discharge line. Do not clamp or restrict the hose braid. The end fittings of these hose assemblies are not intended for reuse. For procurement, this is a critical application item.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title number and date of this specification.

b. Manufacturer's military standard part number or FSCM number, if available.

c. When first article testing is required (see 3.2), or waived (see 4.3.1).

6.3 Subject term (key word listing).

Dynamic beam seal
Aerospace fluid systems

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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MIL-DTL-8815
FILTER

FIGURE 3. Test setup for pneumatic surge test.

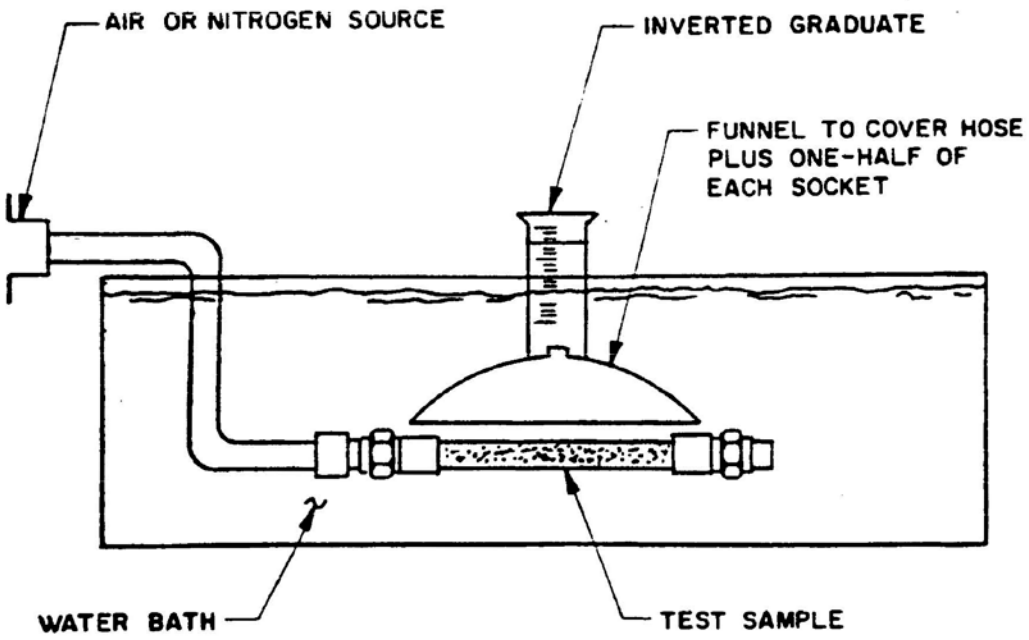


FIGURE 4. Test setup for pneumatic effusion test.

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CONCLUDING MATERIAL

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